SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_fourgee_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	fourgee
Project / Solution Name	Oroboros: Consume. Regrow. Plan.
Name (Leader)	KHAIRUL ZARIFF ADEEB BIN KHAIRUL ZIAD
Email & Phone (Leader)	zariffadeeb@gmail.com / +7 7001027071
University Affiliation(s)	Universiti Teknologi MARA
Live Prototype URL	https://romaaamaker.github.io/oroboros/
Code Repository URL (public / access granted to judges)	https://github.com/romaaamaker/oroboros — public team submission fork. Original commit history is retained. Public access is verified; designated committee collaborator invitations could not be verified.

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Oroboros has a static HTML/CSS/JavaScript frontend hosted on GitHub Pages. Browser code reads generated JSON for occupations, exposure labels, momentum, courses, personas and SDG mappings, then computes scenario scores and learning recommendations. The documented Python pipeline retrieves public data, uses DuckDB and exports static files; dependencies include HTTPX and openpyxl. Repository documentation describes nightly GitHub Actions refreshes; successful scheduled runs have not been verified in this review. The repository also documents an optional FastAPI/Qwen integration, but the public demo labels its LLM as not connected. The team confirms that RunPod was not used and no GPU workload was run. Replit was used for documentation, video-script assistance and the completed pitch deck; it does not host the current prototype. See Appendix A for the active and optional paths.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
HTML5, CSS3, JavaScript	Front-end app and in-browser calculations	Open web standards
Python 3 (exact runtime version not recorded)	Data preparation and validation scripts	PSF License
DuckDB >=1.1; HTTPX >=0.27; openpyxl >=3.1	Declared pipeline dependencies: analytical storage, HTTP data retrieval and spreadsheet ingestion.	MIT / BSD-3-Clause / MIT respectively (verify installed versions and notices before signing)

Name & Version	Purpose in Project	License
Vanilla HTML/CSS/JavaScript; Syne, Figtree and JetBrains Mono via Google Fonts. Optional server: FastAPI >=0.115, Uvicorn[standard] >=0.30, HTTPX >=0.27, Pydantic >=2.7.	Browser presentation and typography; optional API server/validation code, not live model inference.	Fonts: SIL Open Font License. FastAPI/Pydantic: MIT; Uvicorn/HTTPX: BSD-3-Clause. Review transitive dependencies separately.

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
arXiv API (arXiv, Cornell University)	Monthly paper counts per AI capability area (momentum signal)	Free
Hacker News Algolia API (Algolia / Y Combinator)	Counts of hiring-thread mentions per capability area	Free
GitHub REST API (GitHub)	Activity on reference open-source projects	Free
Remotive public API (Remotive)	Job-posting counts (jobs feed)	Free
Arbeitnow API (Arbeitnow)	Job-posting counts (jobs feed)	Free
Microsoft Learn Catalog API (Microsoft)	Course catalogue for learning plans	Free
UN Stats SDG API (United Nations)	SDG target text and Malaysia context	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
GitHub Pages (GitHub)	Hosting of the static prototype	Free plan / global CDN
GitHub Actions (GitHub)	GitHub Pages deployment runs verified successful; nightly refresh configured, but scheduled executions on this fork not verified.	Free plan

Service & Provider	Purpose in Project	Plan / Region
Replit	Documentation, video-script preparation and the completed nine-slide pitch deck. Not the app host.	Account plan/region not recorded; no sponsor-credit claim made.

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
None in the live prototype. Qwen2.5-7B-Instruct is optional only.	Qwen team, Alibaba Cloud (open weights, Hugging Face)	No live model inference, training or fine-tuning claimed. RunPod not used.	Apache-2.0 as declared in repository LICENSE_NOTICES.md.

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
O*NET 31.0 Database	onetcenter.org/database.html	Public	CC BY 4.0	N
GPTs are GPTs (exposure labels)	github.com/openai/GPTs-are-GPTs	Public	MIT; upstream LICENSE checked. Copyright (c) 2024 OpenAI. Include its full notice with redistributed material.	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Anthropic Economic Index	huggingface.co/datasets/Anthropic/EconomicIndex	Public	CC-BY for data; MIT for code, per upstream dataset README. CC-BY version is not specified there; metadata still says MIT. Preserve attribution and verify release-specific terms.	N
Felten et al. AIOE	github.com/AIOE-Data/AIOE	Public	No explicit license grant found upstream. Authors request citation, but redistribution permission remains unresolved; attribution alone is not permission.	N
Signal snapshots (arXiv, HN, GitHub, Remotive, Arbeitnow counts)	Public APIs listed in 3.1	Public (aggregate counts)	Provider API terms	N
UN SDG target text; Microsoft Learn catalogue	sdgs.un.org; learn.microsoft.com	Public	UN and Microsoft terms	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Team-curated SDG map, course list, 5 persona profiles	Repository data/ and pipeline/config/ (per repository map)	Self-collected; personas synthetic	Apache-2.0 for team-owned material when the accompanying LICENSE is uploaded. Third-party datasets retain their own terms.	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Anthropic); exact version/model not confirmed	Development assistance. Exact model and scope were not recorded by the team; no precise version is claimed.	Major (working estimate; final team review required)
Higgsfield AI; exact generation models not confirmed by team	Web-app design. Repository notices also attribute hero imagery/video to Higgsfield; model-level details require team confirmation.	Minor (working estimate; final team review required)
Replit Agent; underlying model not specified	Review and revision of submission documents; timed pitch/demo script; creation and PDF export of the nine-slide pitch deck.	Moderate for documentation, script and pitch-deck work

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Replit Agent compared these drafts with the public app, repository README, dependency manifests and license notices, correcting stack discrepancies and separating optional features from live claims. The team must confirm its original code/data review process, executed tests and results before signing. This review did not execute the application test suite or verify every dataset record, course link or license. Team members retain responsibility for accuracy, attribution and all submitted material.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

Replit platform used: Yes. RunPod used: No (administration exception reported by team). Dedicated sponsor credits or account provisioning: not established; no such credit claim is made.

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit / RunPod	Replit platform used for documentation, script assistance and the completed pitch deck. RunPod: not used. No sponsor credits or GPU workload claimed.	Replit supports documentation, script preparation and the completed pitch deck; not current app hosting or inference. RunPod was not used because the project has no GPU workload. The team reports that the administration confirmed RunPod is not required for this project.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

The browser-based prototype does not depend on Replit or RunPod for runtime computation. Replit supported documentation, script preparation and the completed presentation. RunPod was not used because the project has no GPU workload. The team reports that the administration confirmed RunPod is not required for this project.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The public demo states that the career engine processes user inputs in the browser and uses synthetic personas; its LLM is labelled not connected. This review has not independently tested network traffic, browser persistence or all analytics behaviour, so absolute no-collection/no-retention claims require team verification. The page loads Google Fonts, and hosting/font providers receive ordinary web requests subject to their own policies. Do not enter real CVs in the recorded demo. If the optional API is enabled, pasted text may be sent to the configured FastAPI/model service; the team must document consent, retention, logging and provider terms before claiming local-only processing.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

O*NET is US-based and needs validation for Malaysian work contexts. Exposure measures are not predictions of job loss. Momentum caps, scenario multipliers and classification thresholds are modelling assumptions. Public feeds can fail; documentation describes cached snapshots, but successful refresh operation needs verification. The optional language-model path may introduce hallucination, privacy and provider risks; it is not connected in the public demo. AIOE licensing remains unresolved in repository notices. No measured user-impact results are established in this review. Confirm course availability, local relevance and accessibility before final submission.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☐ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☐ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☐ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☐ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☐ All fields in this template have been completed within the word limits stated.

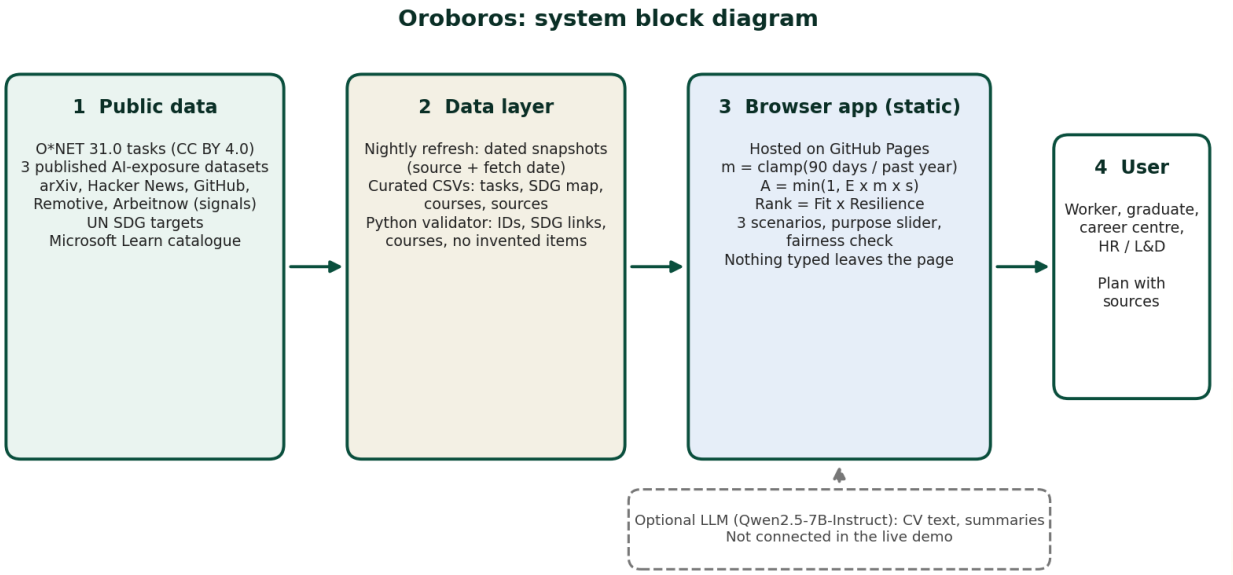
8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team’s knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



25 October 2026

Appendix A: System block diagram



ACTIVE: Public occupational/research/API data → documented Python + DuckDB pipeline → dated static JSON → GitHub Pages → user browser → task scenarios, role rankings and learning plan.
OPTIONAL / NOT ACTIVE IN PUBLIC DEMO: Browser → FastAPI validation → Qwen model endpoint → validated response. RunPod not used; no GPU inference claimed.
SUBMISSION SUPPORT: Claude Code → development assistance; Higgsfield AI → web-app design; Replit Agent → documentation and pitch/demo script. The nine-slide Replit pitch deck is complete and exported to PDF.